

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

SE 19-206-0306 PRINCIPLES OF SAFETY MANAGEMENT

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Explain the concepts of safety and safety equipments and safety organization.

CO2: Analyse job safety, accidents through accident models, estimate costs.

CO3: Estimate safety performance using various indices.

CO4: Produce different types of permits, reports, and compliance check lists.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Describe the functions of Safety Committees.	3	L1	1	1
(b) List any five accident causation theory and explain any one.	3	L2	1	1
(c) What are the advantages of behaviour oriented training?	3	L2	2	1
(d) How will you diminish accidents through good house keeping?	3	L1	2	1
(e) Discuss the use and significance of "plant safety inspection".	3	L1	3	2
(f) Discuss the difference between safety audit and safety survey.	3	L1	3	1
(g) "Describe the data collection methods used in accident investigation.	3	L2	4	2
(h) Accident investigation is a "fact finding process" not a "fault finding process" comment on this statement.	3	L1	4	2

PART B

(4 × 12 = 48)

II. (a) Explain the major factors responsible for 98% of accidents.	6	L1	1	1
(b) Discuss the duties of a safety officer in a petroleum industry.	6	L2	1	2
OR				
III. Discuss the salient features of any three theory of accident causation.	12	L2	1	2
IV. (a) Design a work permit system for hot work.	8	L1	2	2
(b) Discuss the role of effective communication in safety.	4	L3	2	2
OR				
V. (a) Explain different off the job training methods.	8	L2	2	1
(b) Discuss the 5S concept of house keeping in accident prevention.	4	L1	2	1

(P.T.O.)

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VI.

Calculate the frequency, severity, incidence rates and FSI of the factory based on the data given in the table for three successive calendar years and comment on the safety performance.

Marks 12 BL L4 CO 3 PO 3

Year	2020	2021	2022
Number of deaths	1	2	0
Number of disabling injuries other than deaths	12	15	11
Man-days lost due to disabling injuries (excluding death)	400	800	600
Average number of workers	3000	4500	200
Total man-hours worked in the year	8,00,000	10,00,000	7,00,000

OR

- VII. (a) Prepare a sample job safety analysis work sheet for any hazardous job. 6 L2 3 2
- (b) Describe the different types of PPE used for electrical work. Discuss the limitations, if any associated with them. 6 L2 3 1

- VIII. Describe the various steps involved in accident investigation. Prepare a sample report. 12 L1 4 1

OR

- IX. (a) Discuss change analysis with a sample worksheet. 8 L3 4 2
- (b) Construct a fault tree diagram illustrating a typical failure process. 4 L1 4 1

Blooms's Taxonomy Levels

L1 – 44.16%, L2 – 34.16%, L3 – 16.67%, L4 – 5%.
